



NYU

SCHOOL OF GLOBAL
PUBLIC HEALTH

Lecture 4: Causal Inference

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How do we study “causes”?

(1) Theory of Causation

- *Research Question: Exposure to certain risk factors causes disease*

(2) Formulate a testable hypothesis

- Exposure to variable X will be ((related)) to disease Y

(3) Design and Conduct a Study

- *Randomized Trials, Cohort Studies, Case-Control Studies, Ecological Studies*
- *Goal of the study is to Minimize Bias, Minimize Random Error, Collect Data on Confounding Variables*

(4) Analyze the data

(5) Interpret the results

Bradford-Hill Criteria:
Temporality and
Biological Plausibility

Bradford-Hill Criteria:
Strength of Association
Dose Response
Consideration of alternate explanations
Cessation of Exposure
Coherence with Established facts
Specificity of Association
Replication of Findings

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Causal Paradigms - Introduction

- Very rarely* is one factor a sufficient cause of disease

Exposure to Risk Factor → *Disease*

- *genetic diseases can be exceptions

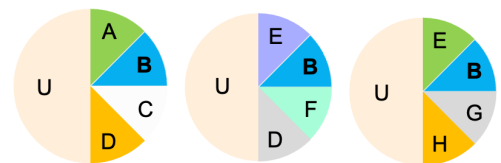
Defect in BOTH Hex-A genes → *Tay-Sachs*

- More often diseases are the result of a complex system of many related factors (multiple causes)
- But we need frameworks for understanding these connections.

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Causal Paradigm – Sufficient & Necessary Causes

- **Sufficient Cause:** Minimum set of conditions for an outcome to occur
 - May be more than one for each disease or condition
 - Possible that every case of outcome has a unique pie
 - The “U” component is often largest - for many diseases, we don’t fully understand disease causation
 - If we could specify the mechanisms perfectly, could we predict all disease?
 - Maybe....BUT there are likely infinite combinations of these causal pies
- In risk factor epidemiology, we focus on one component cause and often ignore the complement causes within a sufficient cause. *Is this okay?*



Rothman KJ et al. Modern Epidemiology, 3rd ed. 2008.

- **A, B, C, & D** = component causes
- **U** = unknown cause(s)
- **B** = necessary cause

QUESTION: What happens if you eliminate one component cause?
- A? B?

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Component Causes as “Exposures”

- Component causes are exposures we are interested in studying
 - ***Does smoking cause lung cancer?***
 - “Smoking” is too imprecise.. How much? How long? What type? Age of onset?
 - Infinite combinations of above attributes, each may have a different risk
 - Ignoring these attributes → treat all exposures (from 1 lifetime cigarette to a 10 pk/day) as exposed...*How might this impact the results?*
- Exposures have distinct attributes:
 - Dose
 - Duration
 - Induction period (NOT LATENCY)
- Specifying attributes improves the study’s ability to identify a potential causal effect

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Measuring exposures: Dose attributes

- Time weighted average dose
 - Grams of fat per day
- Maximum dose
 - Highest adult body weight
- Body weight or surface area scaled
 - Grams of alcohol per kilogram body weight
- Cumulative dose
 - Pack years of cigarettes

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Measuring Exposures: Duration attributes

- Total time of exposure
 - years employed
- Biologically relevant time of exposure
 - Smoking before first pregnancy
- Time of exposure beyond a minimum
 - Years of driving after age 25
- Time of exposure after gathering another component cause
 - HIV infection after HPV infection

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Induction period attributes

- Time between completion of a COMPONENT cause (*i.e. exposure of interest*) and completion of the SUFFICIENT cause (*i.e. disease occurrence*)
 - Induction period doesn't characterize disease
 - Characterizes component cause-disease pair
 - Every disease has a component cause with zero induction time
 - Failure to exclude induction time from person time biases towards the null

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Strength is Determined by Complements

- Strength of a risk factor
 - Typically measured on the relative scale
- Is determined by the relative prevalence in the population of the causal complements
 - Also affected by the competing risks of other sufficient causes for the same disease

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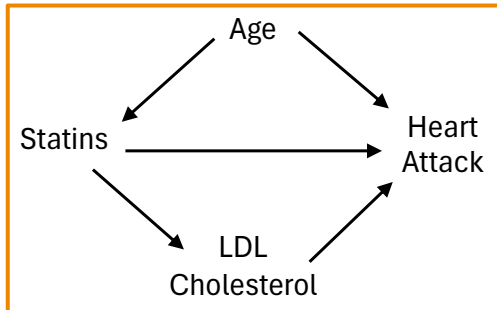
Causal diagrams

- Directed acyclic graphs (DAGs) are often used to represent causal concepts in epidemiology
- Breaking that down:
 - Graph: nodes (labeled as variables); lines connecting nodes.
 - Directed: lines have a single arrowhead
 - Acyclic: no loops
 - Advanced note: of course, feedback loops can exist. For example, sick persons are more likely to take medicine as prescribed whereas, those who are sick and less symptomatic may be less likely to take meds as prescribed.

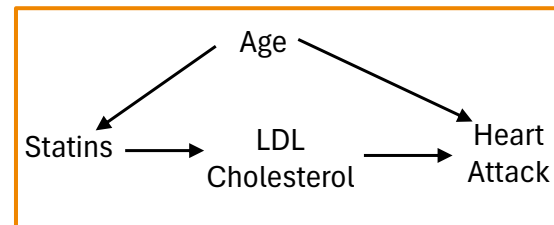
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Example

- Does use of statins affect risk of heart attack?
 - We'll consider only statins, heart attack, age, and LDL cholesterol levels.



OR



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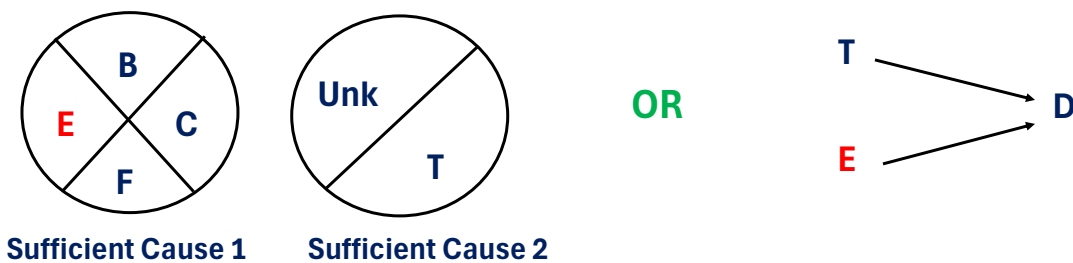
Notes on DAGs

- The data we encode in the DAG helps guide our analysis
 - We construct the DAG based on subject-matter background knowledge/beliefs.
 - Antecedents of both exposure and outcome must be on the DAG (even if we did not collect data on those antecedents)
 - We build DAG **before** doing any analyses and **before** collecting any data
 - Don't make mistake of thinking you construct the DAG from only the observed or collected data.
- DAGs are non-parametric:
 - We use the same DAG for risk differences, risk ratios, rate ratios, etc.
- The arrows mean “(hypothetical) causes”
 - An arrow from Statins to Heart Attack means “if your statin use changes, your MI risk will change.”
 - Arrows are (generally) non-determinative: $E \rightarrow D$ does not mean that if E changes from 0 to 1, then D is deterministically going to change in lockstep.

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Connecting DAGs and causal pies I: Independent risk factors

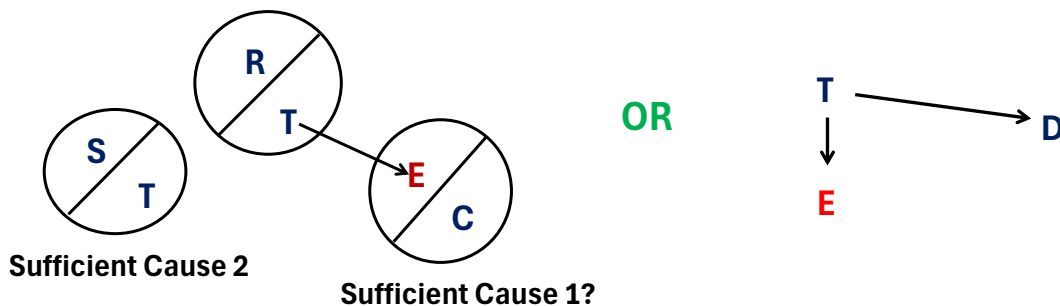
- Say **E** is the cause of interest (exposure) and T is another risk factor
 - “A cause of a disease event is an event, condition, or characteristic that preceded the disease event and without which the disease event would not have occurred at all or would not have occurred until some later time.”
- Denoting independent risk factors for the outcome:



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Connecting DAGs and causal pies II: Confounding

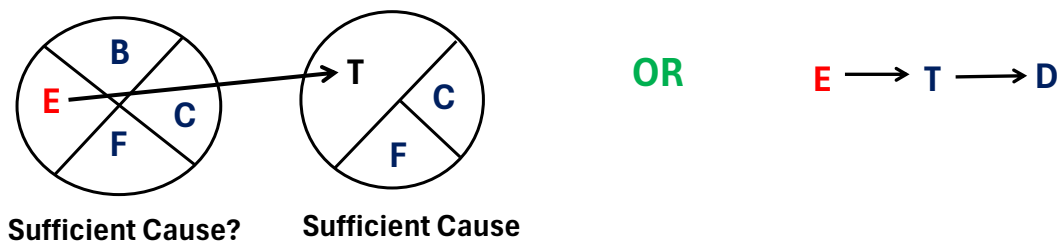
- Say **T** is a confounder of the E (exposure) and D relationship
- Denoting confounding:



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Connecting DAGs and causal pies III: Mediation

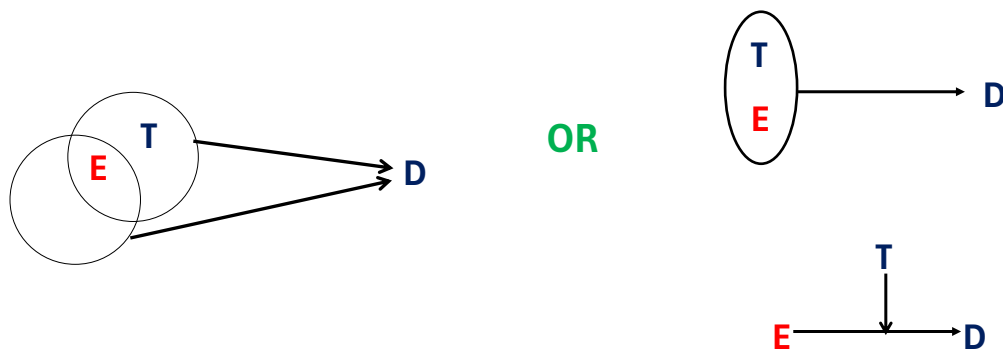
- Say **T** is a mediator of the E (exposure) and D relationship
- Denoting mediation:



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Connecting DAGs and causal pies IV: Effect Measure Modification

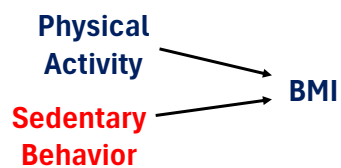
- Say **T** modifies the effect of E (exposure) on D
- Denoting effect measure modification:



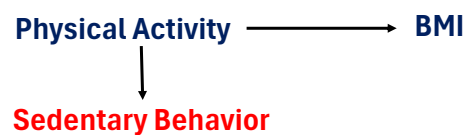
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An example:

1. Independent Risk Factors



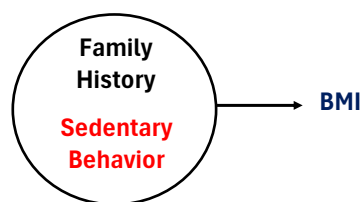
2. Confounding



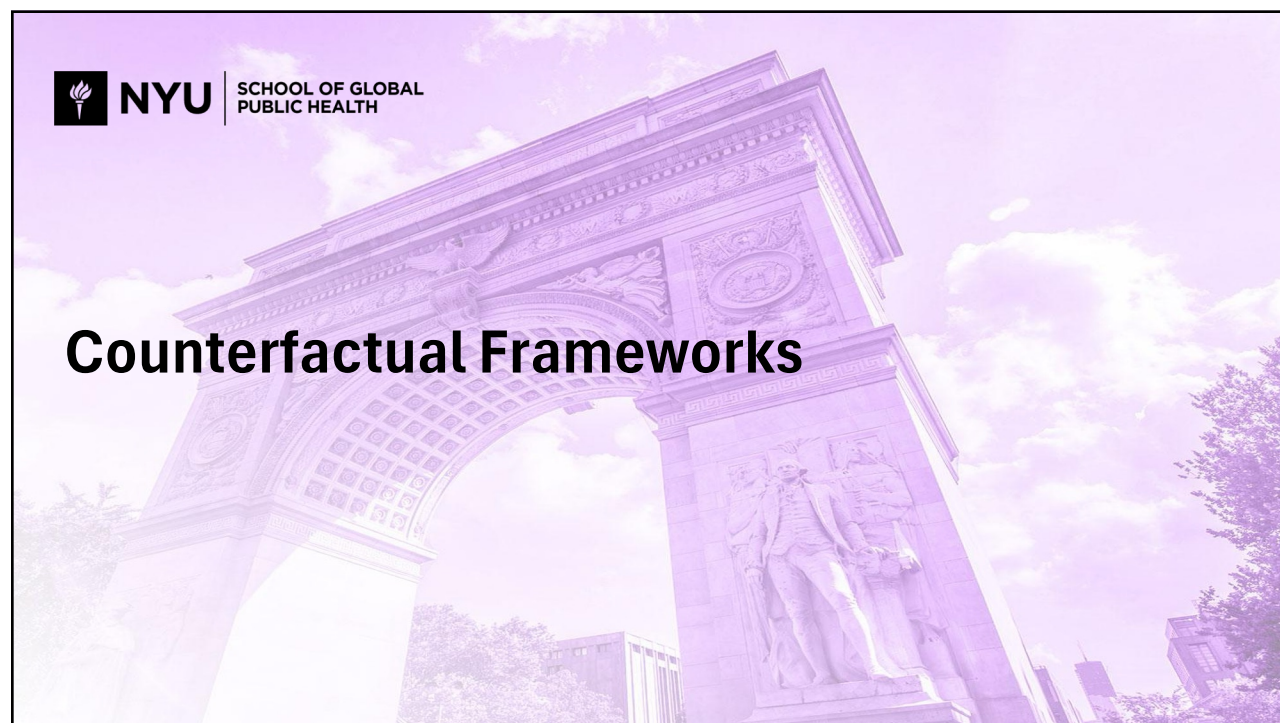
3. Mediation



4. Effect Modification



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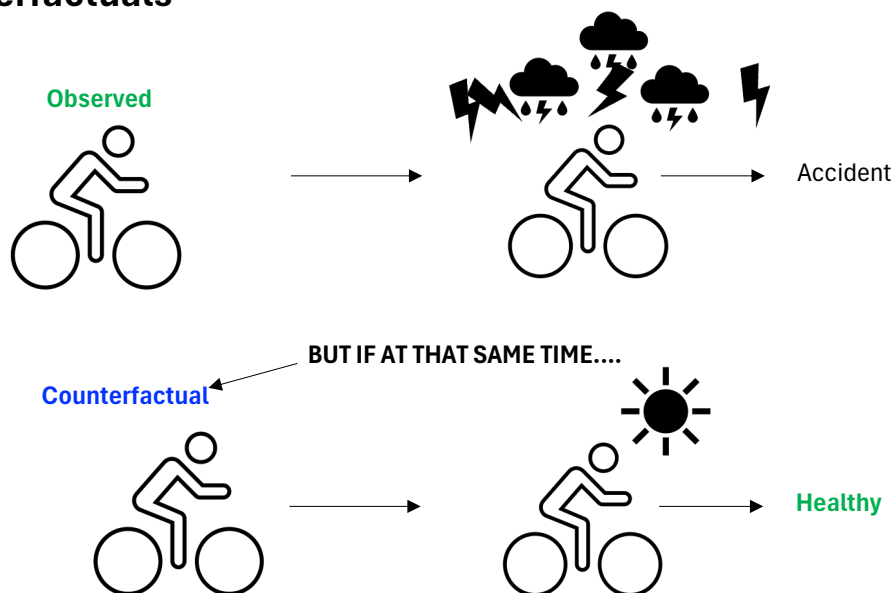
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Empirically evaluating Causality

- In practice we compare a group of exposed subjects with another group of unexposed subjects.
- Thus, the validity of this comparison depends on the *assumption* that the risk of disease in the *unexposed* group is equal to the risk that *would have occurred in the exposed group in the absence of exposure*.
- When this assumption is not true, the observed comparison between exposure groups may not accurately reflect the true difference in risk between the groups (*hint: this is likely due to uncontrolled confounding*)

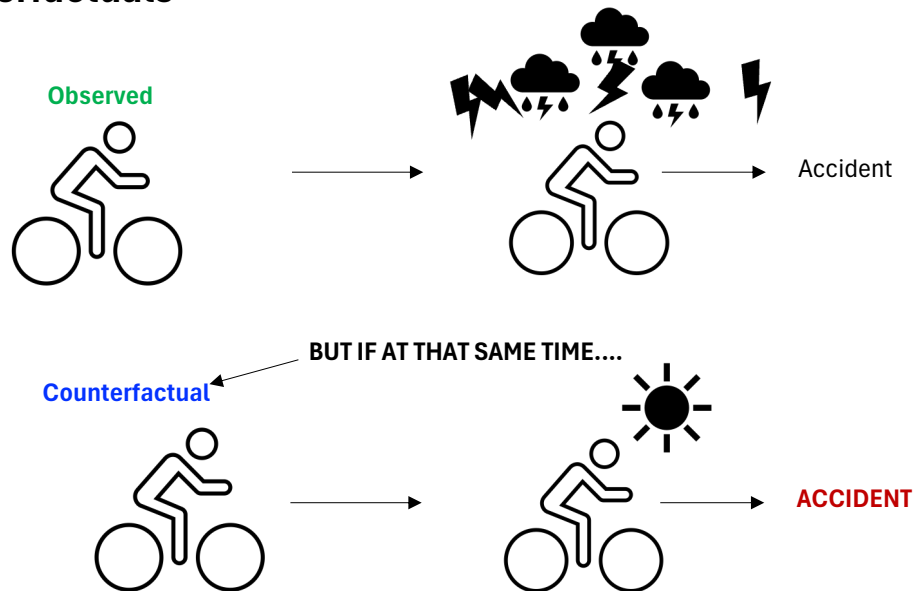
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Counterfactuals



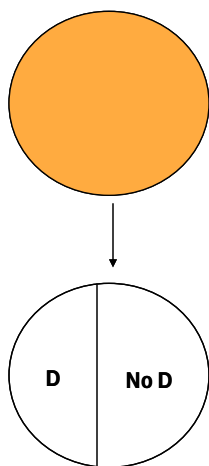
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Counterfactuals



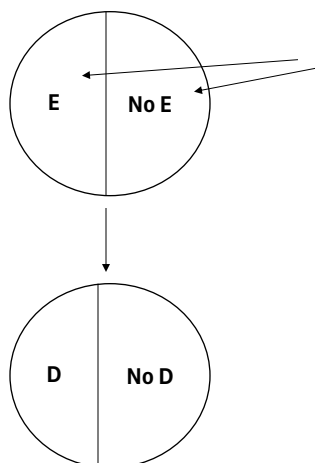
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The counterfactual world



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The actual (real) world



So, the question, how are these two parts different? Are they “exchangeable”? Epidemiology is centrally concerned with ensuring exchangeability or comparability of these two parts of the population

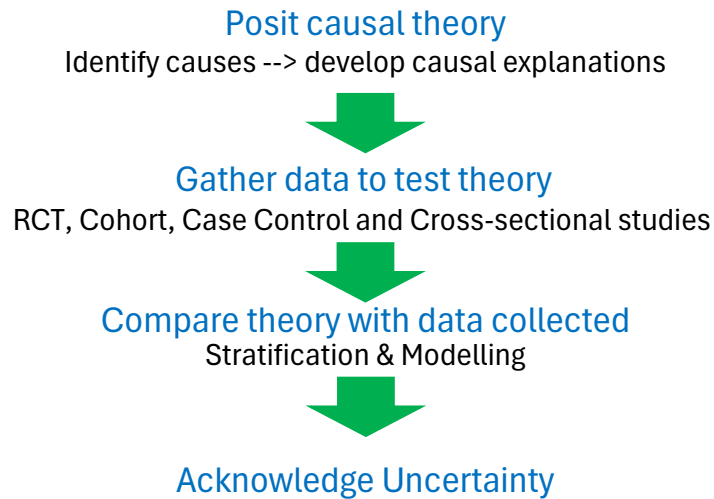
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The “BIG” questions in analytic epidemiology

- Is there a *statistical association* between an exposure and a disease?
- Is it likely to be due to random chance or study bias?
- Is the strength of association large enough to be clinically important?
- Is it consistent with other studies?

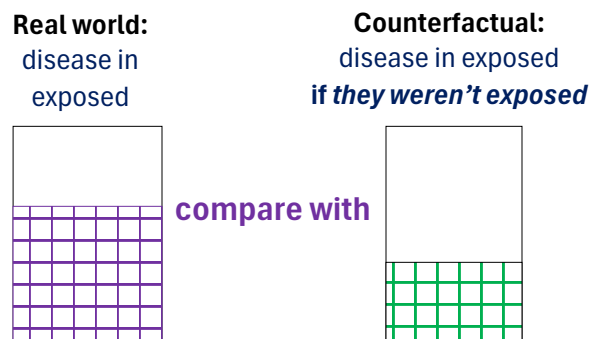
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Goal of epidemiologic methods



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Information we want: Causal Inference



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An elementary model of causal types and their distribution

Causal Type	Individual Disease status if:		Cohort 1
	Exposed	Unexposed	(Exposed)
1) Doomed	1=Disease	1=Disease	p1
2) Exp is Causative	1=Disease	0=No Disease	p2
3) Exp is Preventive	0=No Disease	1=Disease	p3
4) Immune	0=No Disease	0=No Disease	p4

Causal risk difference in cohort 1: $(p1+p2) - (p1+p3) = p2 - p3$
Exposed & develop disease *Had they been unexposed & develop disease*

Causal risk ratio in cohort 1: $\frac{(p1+p2)}{(p1+p3)}$

Causal odds ratio in cohort 1: $\frac{(p1+p2) / (p3+p4)}{(p1+p3) / (p2+p4)}$

NOTE: if $p2 - p3 = 0$ then causal risk and odds ratio = 1
 → bc there is balance between causative and preventative effects

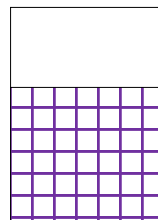
Table 4-1 p 60 ME2 (Rothman and Greenland)

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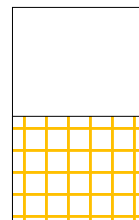
Data we get: Association

Real world:
disease in
exposed

Real world:
A proxy for the counterfactual:
disease in the UNexposed



compare with



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An elementary model of causal types and their distribution in two distinct cohorts

Causal Type	Individual Disease Status		Cohort 1	Cohort 0
	Exposed	Unexposed	(Exposed)	(Unexposed)
1) Doomed	1 = Dx	1 = Dx	p1	q1
2) Causative	1 = Dx	0 = No Dx	p2	q2
3) Preventive	0 = No Dx	1 = Dx	p3	q3
4) Immune	0 = No Dx	0 = No Dx	p4	q4

Causal risk difference: $(p1+p2) - (q1+q3)$
 get disease in cohort 1 (=exposed)   get disease in cohort 0 (=unexposed)

Causal risk ratio: $\frac{(p1+p2)}{(q1+q3)}$

Causal odds ratio: $\frac{(p1+p2) / (p3+p4)}{(q1+q3) / (q2+q4)}$

NOTE: if $q1 + q3 \neq p1 + p3$ then $q1+q3$ cannot be exchanged or substituted for $p1+p3$ and the association measure (risk comparisons) are confounded by the discrepancy between these two quantities

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Risk

- Risk in the exposed (we may know this):

$$\frac{\text{Exposed who get disease by the end of the study period}}{\text{Total number at risk}}$$

- Risk in exposed **had they not been exposed (we cannot know this):**

$$\frac{\text{Exposed who would have disease at end IF they had been unexposed}}{\text{Total number at risk}}$$

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Risk as conceptualized in terms of the unseen “causal” types

- Risk in the exposed is therefore the proportion of types who get disease under exposure condition:

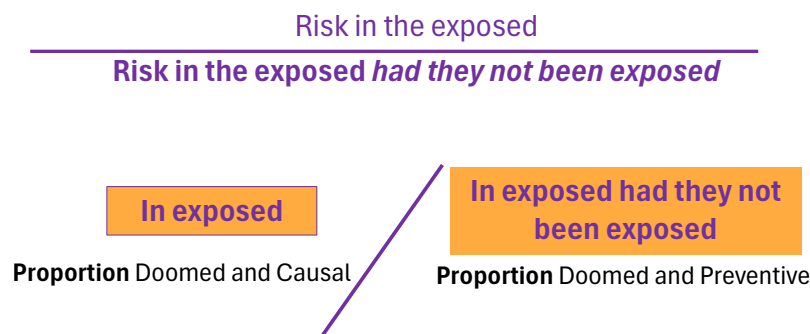
$$\frac{\text{Doomed and causal}}{\text{Doomed, causal, immune, preventive}} = \text{the proportion doomed and causal among the exposed}$$

- Risk in **exposed had they not been exposed** is therefore the proportion of types who get disease if they were not exposed

$$\frac{\text{Doomed and protective}}{\text{Doomed, causal, immune, preventive}} = \text{the proportion doomed and protective among the exposed had they not been exposed}$$

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Causal Contrast (expressed as Risk Ratio)



The causal contrast is the ratio of proportion doomed and causal vs. proportion doomed and preventive **among the same individuals**

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Risk Ratio we CAN calculate with observed data

- Risk in the exposed is the proportion of doomed and causal types among the exposed

$$\frac{\text{Doomed and causal}}{\text{Doomed, causal, immune, preventive}}$$

= the proportion doomed and causal among the exposed

- Risk in unexposed is the proportion of doomed and preventive types among the unexposed

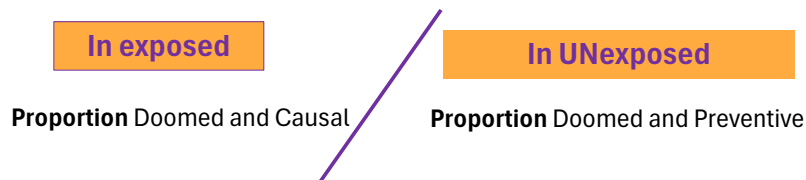
$$\frac{\text{Doomed and preventive}}{\text{Doomed, causal, immune, preventive}}$$

= the proportion doomed and preventive among the UNExposed

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Risk Ratio using substitute for Counterfactual (unexposed people)

$$\frac{\text{Risk in the exposed}}{\text{Risk in the UNExposed}}$$



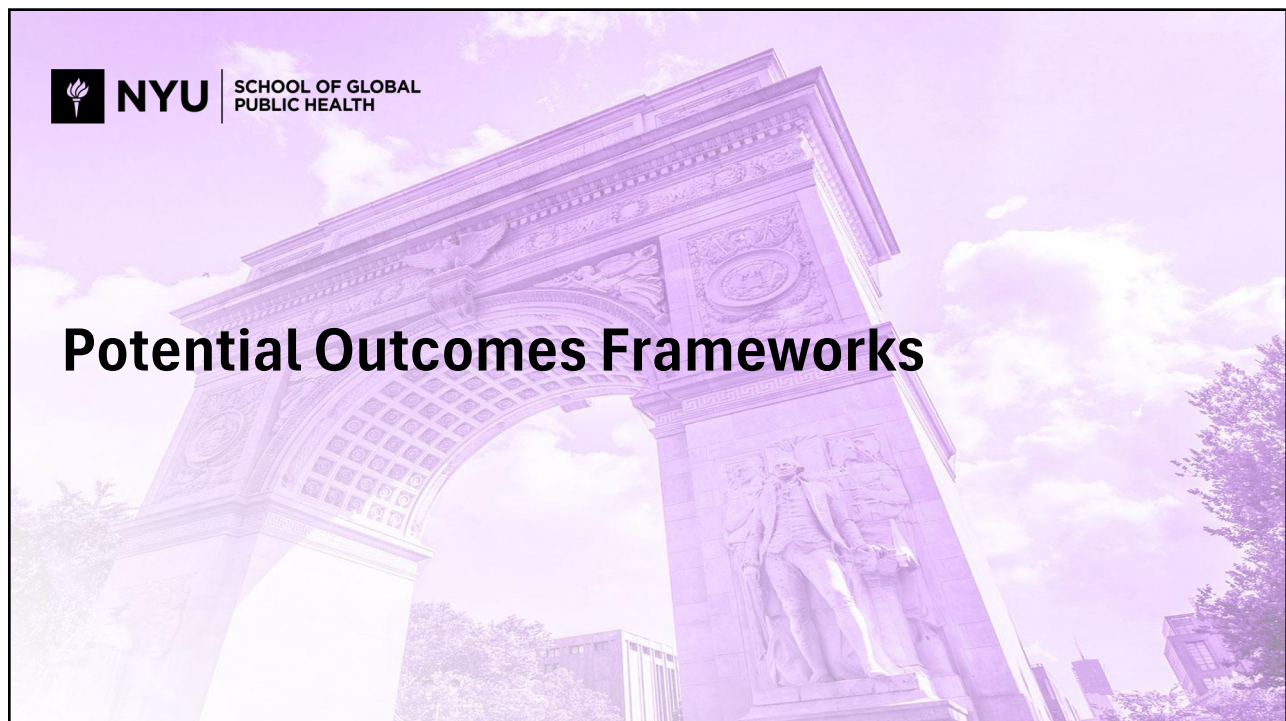
The risk ratio we calculate is the ratio of proportion doomed and causal vs. proportion doomed and preventive **in different groups of people**

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Measures of association – what do we want out of them statistically?

- To help **quantify** the *relationship* between exposures and outcomes
 - Specifically, measures of association are used to quantify the **strength** and **direction** of associations between exposures (E) and outcomes (Y).
- Ultimately, we want to use this information to identify any **effects** of exposures on outcomes

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Potential Outcomes Frameworks

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The “What if....” game

- What if Bruce Wayne’s parents had never been murdered?
- What if that waitress had won the lottery?
- What if you hadn’t gotten your drivers license?

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An example to help us think through causal inference...

- 12 subjects each suffer a headache.
- Some take medicine; others don't.
- One hour later, we ask each of the 12 whether or not the severity of their headache was reduced (did the headache get better)

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Causal Thinking...

Person	X = Medicine (Y/N) 0=no meds/1=meds (Exposure)	Y = Headache 0=headache/1=no headache (Outcome)
1	0	0
2	1	0
3	1	1
4	0	0
5	0	1
6	1	0
7	1	0
8	0	0
9	0	1
10	0	0
11	0	1
12	1	1

- Person 3 took the medicine, and his headache stopped.
- Did the medicine **cause** his headache to stop?
- We don't know.
- To answer this, we need to know what **would** have happened **had they not** taken the medicine.

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Counterfactuals and potential outcomes

- Write Y^0 and Y^1 to represent **the potential outcomes** under both treatments.
- Y^0 = outcome if medicine **NOT** taken.
- Y^1 = outcome if medicine **IS** taken.
- Only one of these is observed: if $X = 0$, Y^0 is observed; if $X = 1$, Y^1 is observed.
 - **The other is counterfactual!**
- But what if we can observe the unobservable. . .

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Causal Thinking...the 'ideal data'

Person	Y^0	Y^1	Causal Effect?
1	0	0	No
2	1	0	Yes, harmful
3	0	1	Yes, protective
4	0	0	No
5	1	1	No
6	0	0	No
7	0	0	No
8	0	0	No
9	1	0	Yes, harmful
10	0	0	No
11	1	1	No
12	0	1	Yes, protective

- An individual-level causal effect is defined for each subject and is given by: $Y^1 - Y^0$

Remember:

Y = No Headache

0=headache/1=no headache (Outcome)

X = Medicine (Y/N)

0=no meds/1=med (Exposure)

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The fundamental problem of causal inference...back to reality

Person	Y^0	Y^1	X	Y
1	0	?	0	0
2	?	0	1	0
3	?	1	1	1
4	0	?	0	0
5	1	?	0	1
6	?	0	1	0
7	?	0	1	0
8	0	?	0	0
9	1	?	0	1
10	0	?	0	0
11	1	?	0	1
12	?	1	1	1

- In reality, we **never** observe **both** Y^0 and Y^1 on the same individual.
- This is sometimes called **the fundamental problem of causal inference**.
- It is therefore over ambitious to try to infer anything about individual-level causal effects.

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Population-level causal effects

- A less ambitious goal is to focus on the **population-level** or **average causal effect**:

$$E_{\text{avg}}(Y^1) - E_{\text{avg}}(Y^0)$$

- Or, since Y (the outcome) is binary:

$$P(Y^1 = 1) - P(Y^0 = 1)$$

- Let's return to the 'ideal' data. . .

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Population-level causal effects (cont.)

Person	Y^0	Y^1	Causal Effect?
1	0	0	No
2	1	0	Yes, harmful
3	0	1	Yes, protective
4	0	0	No
5	1	1	No
6	0	0	No
7	0	0	No
8	0	0	No
9	1	0	Yes, harmful
10	0	0	No
11	1	1	No
12	0	1	Yes, protective

- $P(Y^0 = 1) = 4/12$

- $P(Y^1 = 1) = 4/12$

$$P(Y^1 = 1) - P(Y^0 = 1) = 0$$

- i.e. **no causal effect** at the population level.

Remember:

$Y = \text{Headache}$

$0 = \text{headache}/1 = \text{no headache (Outcome)}$

$X = \text{Medicine (Y/N)}$

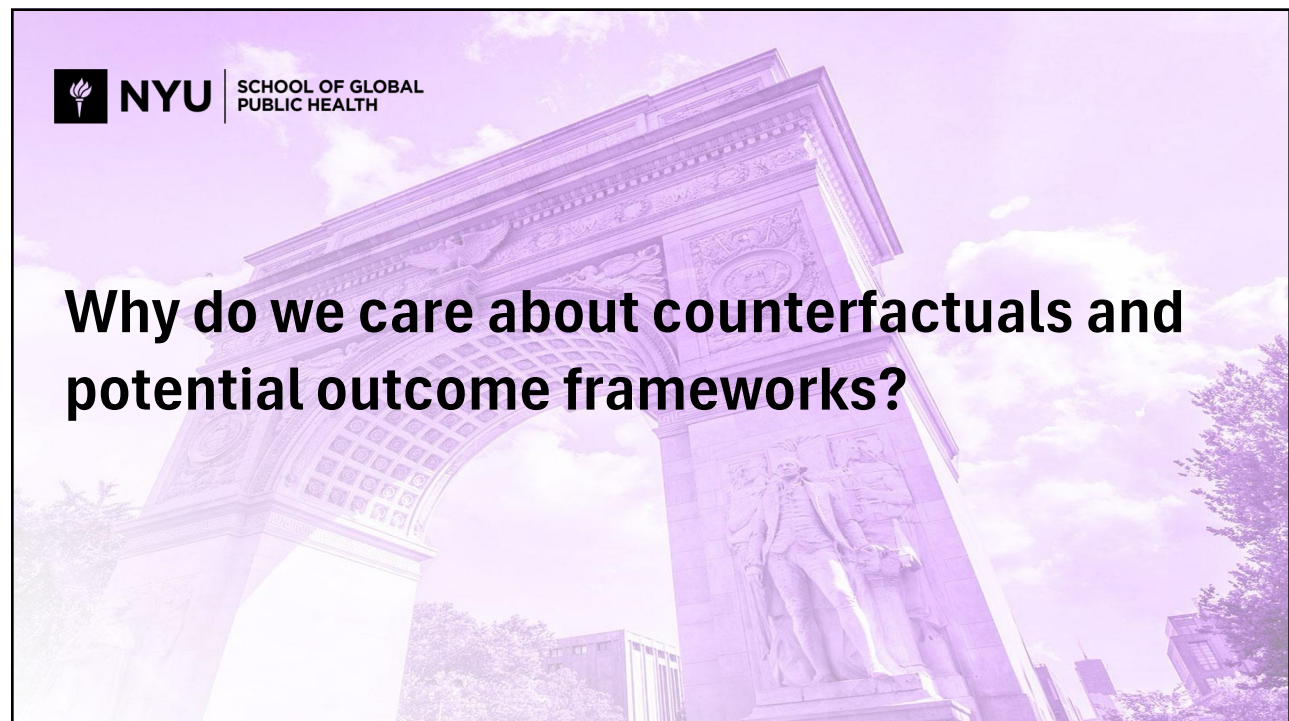
$0 = \text{no meds}/1 = \text{med (Exposure)}$

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Population-level causal effects (cont.)

- We don't know Y^1 for every subject, so we can't simply estimate $P(Y^1 = 1)$ as the proportion of all subjects with $Y^1 = 1$.
- Likewise, we can't simply estimate $P(Y^0 = 1)$ as the proportion of all subjects with $Y^0 = 1$
- Thus, we can't easily estimate $P(Y^1 = 1) - P(Y^0 = 1)$ for the same reason that we can't estimate $Y^1 - Y^0$
 - Causal inference is all about choosing quantities from the observed data (i.e. involving X , Y and other observed variables) that represent reasonable substitutes for hypothetical quantities such as $P(Y^1 = 1) - P(Y^0 = 1)$, which involve unobservable counterfactuals.

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Descriptive/predictive inference and causal inference

- Descriptive/predictive inference – generally deals with either describing a system or predicting (or classifying) an observation.
- Causal inference - explicitly or implicitly involves some sort of intervention that can, at least in principle, be modified

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Descriptive/predictive inference and causal inference - examples

- Descriptive or predictive questions:
 - *Is this group of individuals at high risk of developing side effects due to this medication?*
 - *Is this group of patients suitable for this medication?*
- Causal inference questions:
 - *Which type of anesthetic should these patients receive to minimize the risk of complications during surgery?*
 - *What can be done to reduce the risk of complications during surgery for this group/type of patient?*

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